PROBLEM 42

Particle Attracted to the Origin by $F = m\mu/r^{\alpha}$

Original statement · Problems.pdf, p. 29

Binet + Effective Potential

(Advanced problem.) A point particle P of mass m is attracted by the origin of coordinates O of the inertial reference frame S_0 by a force with magnitude

$$F = \frac{m\mu}{r^{\alpha}}$$

where r is the distance between O and P, and μ, α are known constants.

Notation.
The official text writes $F = m\alpha/r^{\beta}$ (strength α, exponent β) and asks for $\beta = 1, 2, 3, 4$. Here the strength is called μ and the exponent α, so the cases "$\alpha = 1, 2, 3, 4$" below are the official $\beta = 1, 2, 3, 4$, and μ/h^2 is the official α/h^2.

(a) Show that the angular momentum vector and the mechanical energy of P are conserved quantities of motion, and that the motion of P occurs in a plane.

(b) Using polar coordinates (r, θ) in the plane of motion, find a second-order differential equation to solve for $u \equiv 1/r$ as a function of θ (i.e., $u = u(\theta)$).

(c) Find a first integral of this equation and discuss the limits of motion of u as a function of μ/h^2, α, and the values of u and $u' \equiv du/d\theta$ at $t = 0$, where h is the magnitude of the angular momentum vector about the origin per unit mass. For $\alpha = 1, 2, 3$, and 4, determine under which conditions: (i) the particle escapes to infinity ($u \to 0$); (ii) the particle reaches $r = 0$ ($u \to \infty$); (iii) the particle oscillates between two values of u.

Hint: try to solve first the problem with $\alpha = 2$, and then try the other values.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P of mass m attracted to the fixed point O by $\mathbf{F} = -\dfrac{m\mu}{r^\alpha}\mathbf{e}_r$, where $r = \|\overrightarrow{OP}\|$.
- **Hypotheses:** S_0 is inertial; $\mu > 0$ and $\alpha > 0$ are constants; no other force acts; $h \neq 0$ unless stated.
- **Constraints:** none. P is a free particle, so $\mathrm{CDOF} = 3$. Part (a) proves that the motion is planar; that is a consequence of the dynamics, not an imposed constraint. In the plane, $\mathrm{CDOF} = 2$.
- **Coordinates:** polar coordinates (r, θ) in the plane of motion. For the orbit equation we also use $u(\theta) = 1/r(\theta)$ and $u' = du/d\theta$; h is the magnitude of the angular momentum per unit mass.
- **Special case $h = 0$:** the motion is then radial and the Binet formulation does not apply. It is treated at the end of part (c).
- **Notation:** the official text writes $F = m\alpha/r^\beta$ and asks for $\beta = 1, 2, 3, 4$. Here the strength is μ and the exponent is α.

Symbol	Meaning	SI unit
μ	force strength per unit mass, $F/m = \mu/r^\alpha$	$\mathrm{m}^{\alpha+1}/\mathrm{s}^2$
h	specific angular momentum	m^2/s
u	reciprocal radius	m^{-1}
ξ	specific mechanical energy	$\mathrm{J/kg} = \mathrm{m}^2/\mathrm{s}^2$

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with its origin at the attracting centre.
- **Polar basis.** Take $\mathbf{k}$ along $\mathbf{H}_O$ (justified in 3.3). The polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

- **Angular velocity and angular acceleration of $\mathcal{B}_1$ with respect to S_0: $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$** and $\boldsymbol{\alpha}_{10} = \ddot\theta\,\mathbf{k}$.
- **Time derivatives of the unit vectors (Poisson's formula $\dot{\mathbf{e}} = \boldsymbol{\omega} \times \mathbf{e}$):**

$$\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{k} \times \mathbf{e}_r = \dot\theta\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = \dot\theta\,\mathbf{k} \times \mathbf{e}_\theta = -\dot\theta\,\mathbf{e}_r$$

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the force acts along the line OP, so the position vector is simply $r\mathbf{e}_r$. The basis vector $\mathbf{e}_r$ is not constant in S_0, so each derivative uses the product rule and Poisson's formula. These planar polar formulas are valid for a motion contained in the plane of the basis; that the motion is indeed planar is proved in 3.3, and here it only fixes the choice of $\mathbf{k}$.

$$\mathbf{r} = r\,\mathbf{e}_r$$

$$\mathbf{v} = \frac{d}{dt}\left(r\mathbf{e}_r\right) = \dot{r}\,\mathbf{e}_r + r\,\dot{\mathbf{e}}_r = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta$$

$$\mathbf{a} = \ddot{r}\,\mathbf{e}_r + \dot{r}\,\dot{\mathbf{e}}_r + \left(\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta + r\dot{\theta}\,\dot{\mathbf{e}}_\theta = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\mathbf{e}_\theta$$

- The angular momentum per unit mass about O follows from the velocity:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = r\mathbf{e}_r \times \left(\dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta\right) = r^2\dot{\theta}\,\mathbf{k}, \qquad h = r^2\dot{\theta}$$

3.2 Forces

- **Applied force:** the attraction $\mathbf{F} = -\dfrac{m\mu}{r^\alpha}\mathbf{e}_r$. It is central: its line of action always passes through O.
- **Constraint forces:** none. **Inertia forces:** none, because Newton's second law is written in the inertial frame S_0.

> **Pedagogical Justification:** the force is purely radial and depends only on r, so its elementary work $\mathbf{F}\cdot d\mathbf{r} = F(r)\,dr$ is the exact differential of a function of r (equivalently, $\nabla \times \mathbf{F} = \mathbf{0}$). The force is therefore conservative. The potential energy per unit mass is minus the work per unit mass from a reference radius to r (Barrow's rule). For $\alpha > 1$ the reference is at infinity, and the term at the lower limit $s \to \infty$ vanishes because $1 - \alpha < 0$:

$$V_0(r) = -\int_\infty^r \left(-\frac{\mu}{s^\alpha}\right) ds = \mu\left[\frac{s^{1-\alpha}}{1-\alpha}\right]_\infty^r = \mu\left[\frac{r^{1-\alpha}}{1-\alpha} - \lim_{s\to\infty}\frac{s^{1-\alpha}}{1-\alpha}\right] = -\frac{\mu}{(\alpha-1)\,r^{\alpha-1}}$$

For $\alpha \leq 1$ the integral from infinity diverges, so the reference radius r_* is arbitrary. For $\alpha = 1$:

$$V_0(r) = -\int_{r_*}^{r} \left(-\frac{\mu}{s}\right) ds = \mu \left[\ln s\right]_{r_*}^{r} = \mu \ln \frac{r}{r_*}$$

- In all cases the force per unit mass is recovered by differentiating, e.g. for $\alpha > 1$, $V_0'(r) = -\dfrac{\mu}{\alpha - 1} \cdot \left(-(\alpha - 1)\, r^{-\alpha}\right) = \dfrac{\mu}{r^{\alpha}}$, and for $\alpha = 1$, $V_0'(r) = \mu / r$.

3.3 Equations of Motion

> **Pedagogical Justification:** the torque about the fixed point O vanishes because $\mathbf{F}$ is parallel to $\mathbf{r}$. The angular momentum theorem about a fixed point (Notes Eq. 4.53) is a consequence of Newton's second law and gives a first integral right away.

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = r\mathbf{e}_r \times \left(-\frac{m\mu}{r^{\alpha}}\mathbf{e}_r\right) = \mathbf{0} \implies \left.\frac{d\mathbf{H}_O}{dt}\right|_0 = \mathbf{0} \implies \mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v} = \text{const}$$

- **Planar motion.** From $\mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v}$, both $\mathbf{r} \cdot \mathbf{H}_O \equiv 0$ and $\mathbf{v} \cdot \mathbf{H}_O \equiv 0$. Then P stays in the plane through O perpendicular to $\mathbf{H}_O$, and we orient $\mathbf{k}$ along $\mathbf{H}_O$.
- **Newton's second law in the plane.** With $\mathbf{a}$ from 3.1 and $\mathbf{F}$ from 3.2, projected on $\mathbf{e}_r$ and $\mathbf{e}_\theta$:

$$m\left(\ddot{r} - r\dot{\theta}^2\right) = -\frac{m\mu}{r^{\alpha}}, \qquad m\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) = 0$$

- **Check of closure:** two unknowns, $r(t)$ and $\theta(t)$, and two independent scalar equations.
- **Angular momentum from the transverse equation.** Multiplying by r/m and using the product rule:

$$\frac{d}{dt}\left(r^2\dot{\theta}\right) = 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} = r\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) = 0 \implies h = r^2\dot{\theta} = \text{const}$$

- **Energy from the radial equation.** The force is conservative (3.2), so the mechanical energy per unit mass $\xi = \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + V_0(r)$ is a first integral. Differentiating with the product and chain rules, and using $V_0'(r) = \mu / r^{\alpha}$:

$$\dot{\xi} = \dot{r}\ddot{r} + r\dot{r}\dot{\theta}^2 + r^2\dot{\theta}\ddot{\theta} + V_0'(r)\dot{r} = \dot{r}\left(\ddot{r} - r\dot{\theta}^2\right) + r\dot{\theta}\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) + \frac{\mu}{r^\alpha}\dot{r}$$

$$\dot{\xi} = \dot{r}\left(-\frac{\mu}{r^\alpha}\right) + 0 + \frac{\mu}{r^\alpha}\dot{r} = 0 \implies \xi = \tfrac{1}{2}v^2 + V_0(r) = \text{const}$$

3.4 Integration and Answers

(a) Conserved quantities and planar motion

- The angular momentum $\mathbf{H}_O = m\,\mathbf{r}\times\mathbf{v}$ and the mechanical energy $\xi = \frac{1}{2}v^2 + V_0(r)$ are conserved, and the motion occurs in the plane through O perpendicular to $\mathbf{H}_O$. All three statements were proved in 3.3.

(b) Binet equation for a power law

> **Pedagogical Justification:** we want the orbit $u(\theta)$, so time is eliminated with $h = r^2\dot{\theta}$, i.e. $\dot{\theta} = hu^2$. The chain rule for $r = u^{-1}$ gives $dr/d\theta = -u'/u^2$.

$$\dot{r} = \frac{dr}{d\theta}\dot{\theta} = \left(-\frac{u'}{u^2}\right)\left(hu^2\right) = -hu', \qquad \ddot{r} = \frac{d}{dt}\left(-hu'\right) = -hu''\,\dot{\theta} = -h^2u^2u''$$

- The radial equation $\ddot{r} - r\dot{\theta}^2 = -\mu/r^\alpha = -\mu u^\alpha$ with $r\dot{\theta}^2 = \frac{1}{u}\left(hu^2\right)^2 = h^2u^3$ becomes

$$-h^2u^2u'' - h^2u^3 = -\mu u^\alpha \quad \overset{\div\,(-h^2u^2)}{\Longrightarrow} \quad \boxed{u'' + u = \frac{\mu}{h^2}\,u^{\alpha-2}}$$

(c) First integral and limits of the motion

> **Pedagogical Justification:** write the specific energy with $\dot{r} = -hu'$ and $r\dot{\theta} = hu$. Since ξ is constant, the result is a first integral (a quadrature) with no second derivative.

$$\xi = \tfrac{1}{2}h^2\left[(u')^2 + u^2\right] + V_0\left(\frac{1}{u}\right) \implies \boxed{(u')^2 + W_{\text{eff}}(u) = E^*}, \quad E^* \equiv \frac{2\xi}{h^2}, \quad W_{\text{eff}}(u) = u^2 + \frac{2V_0(1/u)}{h^2}$$

- **Derivative of W_{eff} (chain rule).** With $\dfrac{d}{du}\left(\dfrac{1}{u}\right) = -\dfrac{1}{u^2}$ and $V_0'(r) = \mu/r^\alpha$ evaluated at $r = 1/u$:

$$\frac{d}{du}V_0\left(\frac{1}{u}\right) = V_0'\left(\frac{1}{u}\right)\left(-\frac{1}{u^2}\right) = \mu u^\alpha\left(-\frac{1}{u^2}\right) = -\mu u^{\alpha-2} \implies W_{\text{eff}}'(u) = 2u - \frac{2\mu}{h^2}u^{\alpha-2}$$

- **Check.** $\dfrac{d}{d\theta}\left[(u')^2 + W_{\text{eff}}\right] = u'\left(2u'' + W_{\text{eff}}'\right) = 0$. Where $u' \neq 0$ this gives $u'' = -\tfrac{1}{2}W_{\text{eff}}'$, which is the Binet equation; at the points where $u' = 0$ the Binet equation holds by continuity.
- **Limits of the motion.** Because $(u')^2 \geq 0$, motion exists only where $W_{\text{eff}}(u) \leq E^*$. The turning points ($u' = 0$) are the roots of $W_{\text{eff}}(u) = E^*$, and a minimum or a maximum of W_{eff} is a circular orbit. The constant is fixed by the initial data $u_0 = u(0)$ and $u_0' = u'(0)$:

$$E^* = (u_0')^2 + W_{\text{eff}}(u_0)$$

- The three questions are: **(i)** escape, $u \to 0$; **(ii)** reaching the origin, $u \to \infty$; **(iii)** oscillation between two values of u.

Case $\alpha = 2$: inverse square (Kepler)

- Binet: $u'' + u = \mu/h^2$. Potential: $V_0 = -\mu u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2}u = \left(u - \dfrac{\mu}{h^2}\right)^2 - \left(\dfrac{\mu}{h^2}\right)^2$, a parabola with minimum $-(\mu/h^2)^2$ at $u = \mu/h^2$. The constant is $E^* = (u_0')^2 + u_0^2 - \dfrac{2\mu}{h^2}u_0$.
- **(i) Escape:** possible if and only if $E^* \geq W_{\text{eff}}(0) = 0$, i.e. $(u_0')^2 + u_0^2 \geq \dfrac{2\mu}{h^2}u_0$ (parabola or hyperbola). For $E^* > 0$ a particle moving inward ($u_0' > 0$) is first reflected at the positive root and then escapes.

- **(ii) Reaching $r = 0$:** impossible for $h \neq 0$, because $W_{\text{eff}} \to +\infty$ as $u \to \infty$.
- **(iii) Oscillation:** for $-(\mu/h^2)^2 < E^* < 0$: u oscillates between the two roots (periapsis and apoapsis of an ellipse); $E^* = -(\mu/h^2)^2$ is the circular orbit.

Case $\alpha = 1$: $F \propto 1/r$

- Binet: $u'' + u = \dfrac{\mu}{h^2 u}$. With the reference radius r_*, $V_0 = \mu \ln(r/r_*) = -\mu \ln(u r_*)$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2}\ln(u r_*)$ and $E^* = (u_0')^2 + u_0^2 - \dfrac{2\mu}{h^2}\ln(u_0 r_*)$.
- $W_{\text{eff}} \to +\infty$ both as $u \to 0$ (because $-\ln u \to +\infty$) and as $u \to \infty$. Its minimum is at $W_{\text{eff}}' = 2u - \dfrac{2\mu}{h^2 u} = 0$, i.e. $u_c = \sqrt{\mu}/h$.
- **(i)** Escape is impossible for any finite energy. **(ii)** Reaching $r = 0$ is impossible. **(iii)** Every orbit with $E^* > W_{\text{eff}}(u_c)$ oscillates between two finite turning points; $E^* = W_{\text{eff}}(u_c)$ is the circular orbit $r = h/\sqrt{\mu}$. Since $E^* \geq W_{\text{eff}}(u_c)$ always, every motion with $h \neq 0$ is of type (iii) or circular.

Case $\alpha = 3$: $F \propto 1/r^3$

- $V_0 = -\dfrac{\mu}{2r^2} = -\dfrac{\mu}{2}u^2$. Binet: $u'' + \lambda u = 0$ with $\lambda \equiv 1 - \mu/h^2$. Then $W_{\text{eff}} = \lambda u^2$ and $(u')^2 + \lambda u^2 = E^*$.
- **$h^2 > \mu$ ($\lambda = \kappa^2 > 0$, $\kappa = \sqrt{1 - \mu/h^2}$):** $u = A\cos(\kappa\theta + \delta)$ with $A = \sqrt{u_0^2 + (u_0'/\kappa)^2}$. The cosine vanishes at a finite angle, so $u \to 0$ in every case: **(i) the particle always escapes**; a particle moving inward ($u_0' > 0$) is first reflected at the periapsis $u = A$. **(ii)** never, since $u \leq A$. **(iii)** no oscillation between two positive values.
- **$h^2 < \mu$ ($\lambda = -\kappa^2 < 0$, $\kappa = \sqrt{\mu/h^2 - 1}$):** $u = C_1 e^{\kappa\theta} + C_2 e^{-\kappa\theta}$ with $C_1 = \frac{1}{2}(u_0 + u_0'/\kappa)$, $C_2 = \frac{1}{2}(u_0 - u_0'/\kappa)$. If $C_1 > 0$ (i.e. $u_0' > -\kappa u_0$), $u \to \infty$: **(ii) the particle reaches the origin**. If $C_1 < 0$ ($u_0' < -\kappa u_0$), u reaches 0 at a finite angle: **(i) escape**. If $C_1 = 0$, $u = u_0 e^{-\kappa\theta} \to 0$ only asymptotically (spiral $r = r_0 e^{\kappa\theta}$), a limiting case of escape. **(iii)** no oscillation.
- **$h^2 = \mu$ ($\lambda = 0$):** $u'' = 0 \Rightarrow u = u_0 + u_0'\theta$ (reciprocal spiral). $u_0' > 0$: reaches the origin; $u_0' < 0$: escapes at $\theta = u_0/|u_0'|$; $u_0' = 0$: circle $r = 1/u_0$.

Case $\alpha = 4$: $F \propto 1/r^4$

- Binet: $u'' + u = \dfrac{\mu}{h^2}u^2$. $V_0 = -\dfrac{\mu}{3}u^3$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{3h^2}u^3 = u^2\left(1 - \dfrac{2\mu}{3h^2}u\right)$ and $E^* = (u_0')^2 + u_0^2 - \dfrac{2\mu}{3h^2}u_0^3$.
- $W_{\text{eff}}'(u) = 2u - \dfrac{2\mu}{h^2}u^2 = 2u\left(1 - \dfrac{\mu}{h^2}u\right)$: stationary points $u = 0$ (minimum, $W = 0$) and $u_b = h^2/\mu$ (maximum, barrier $W_{\max} = u_b^2 - \dfrac{2\mu}{3h^2}u_b^3 = \dfrac{h^4}{3\mu^2}$). For $u > u_b$, W_{eff} decreases without bound and crosses zero at $u_z = 3h^2/(2\mu) > u_b$.

- **There is no interior minimum**, so the allowed set $\{W_{\text{eff}} \leq E^*\}$ either contains $u = 0$ or extends to $u \to \infty$. Because $\dot{\theta} = hu^2 > 0$, u' has the sign of $\dot{u}$.
- **Allowed regions by energy** (with $u_0 > 0$):
  - $E^* < 0$: only $u \geq u_2$ with $u_2 > u_z$ is allowed.
  - $E^* = 0$: $u = 0$ is a double root, so the allowed set is $\{0\} \cup [u_z, \infty)$; a particle at finite r has $u_0 \geq u_z$.
  - $0 < E^* < W_{\max}$: two roots $u_1 \in (0, u_b)$ and $u_2 > u_b$; allowed $u \leq u_1$ or $u \geq u_2$.
  - $E^* = W_{\max}$: u_b is a stationary point of W_{eff}; the unstable circular orbit $u = u_b$ and the motions that approach it asymptotically.
  - $E^* > W_{\max}$: no turning points.
- **Motion in each region.** For $0 < E^* < W_{\max}$ and $u_0 \leq u_1$: if $u_0' > 0$ the particle is reflected at u_1, and in any case it escapes. For $u_0 \geq u_2$: if $u_0' < 0$ it is reflected at u_2, and in any case it falls to the origin. For $E^* = W_{\max}$: with $u_0 < u_b$, $u_0' < 0$ escapes and $u_0' > 0$ approaches u_b asymptotically; with $u_0 > u_b$, $u_0' > 0$ falls to the origin and $u_0' < 0$ approaches u_b asymptotically. For $E^* > W_{\max}$, $u_0' < 0$ escapes and $u_0' > 0$ falls to the origin.
- **(i)** escape: $0 < E^* < W_{\max}$ with $u_0 < u_b$; or $E^* = W_{\max}$ with $u_0 < u_b$ and $u_0' < 0$; or $E^* > W_{\max}$ with $u_0' < 0$.
- **(ii)** reaching $r = 0$: $E^* \leq 0$ (then $u_0 \geq u_z$); or $0 < E^* < W_{\max}$ with $u_0 > u_b$; or $E^* = W_{\max}$ with $u_0 > u_b$ and $u_0' > 0$; or $E^* > W_{\max}$ with $u_0' > 0$.
- **(iii)** bounded oscillation between two values of u: **does not occur**; it would need an interior minimum of W_{eff}, and the only bounded motion is the unstable circle $u = u_b$.

Special case $h = 0$ (radial motion)

> **Pedagogical Justification:** with $h = 0, \dot{\theta} = 0$ and the particle moves on a straight line through O. The Binet equation does not apply ($u = 1/r$ cannot be expressed as a function of θ), so we use the radial equation and the energy directly: $\ddot{r} = -\mu/r^\alpha < 0$ and $\frac{1}{2}\dot{r}^2 + V_0(r) = \xi$.

- **(ii)** A particle with $\dot{r}_0 < 0$ reaches $r = 0$ for every α, since the acceleration is always inward. **(i)** Escape requires $\dot{r}_0 > 0$ and $\xi \geq \lim_{r\to\infty} V_0 = 0$, which exists only for $\alpha > 1$; for $\alpha \leq 1$, $V_0 \to +\infty$ and the particle always turns back. **(iii)** Without angular momentum, a particle that turns back falls to the origin, so there is no oscillation between two values.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[F/m] = \mathrm{m/s^2} = [\mu]/\mathrm{m}^\alpha$ gives $[\mu] = \mathrm{m}^{\alpha+1}/\mathrm{s}^2$. With $[h^2] = \mathrm{m}^4/\mathrm{s}^2$ and $[u] = \mathrm{m}^{-1}$, $\dfrac{\mu}{h^2}u^{\alpha-2}$ has units $\mathrm{m}^{\alpha-3}\,\mathrm{m}^{-(\alpha-2)} = \mathrm{m}^{-1}$, the same as u and u'' (θ is dimensionless).
- **Stability of circular orbits (consequence of the first integral):** for general α, $W'_{\text{eff}}(u_c) = 0$ gives $u_c^{\alpha-3} = h^2/\mu$ and $W''_{\text{eff}}(u_c) = 2 - 2(\alpha-2)\dfrac{\mu}{h^2}u_c^{\alpha-3} = 2(3-\alpha)$, so the circular orbit is a minimum (stable) for $\alpha < 3$ and a maximum (unstable) for $\alpha > 3$, in agreement with the cases above ($\alpha = 4$: barrier at $u_b = h^2/\mu$).
- **Check against the official key:** `Problems.pdf` has no entry for Problem 42 in its solution keys, so the result is verified by independent derivation. A symbolic check with sympy confirms the substitution $\dot r = -hu'$, $\ddot r = -h^2u^2u''$, the Binet equation, the first integral $\frac{d}{d\theta}\left[(u')^2 + W_{\text{eff}}\right] = 0$, the barrier $W_{\max} = h^4/(3\mu^2)$ and $W''_{\text{eff}} = 2(3-\alpha)$.
- **Numerical check** (Python, DOP853 with relative tolerance 10^{-12} on $u'' = -u + (\mu/h^2)u^{\alpha-2}$, units with $\mu = h = 1$ unless stated). Blow-up angles are given for the cut-off $u = 10^6$; they change by less than 0.003 rad when the cut-off is 10^9.

Case	Initial data (u_0, u'_0)	Predicted	Integration result
$\alpha = 2$	$(1.5, 0)$	bounded, 0.5 to 1.5	$u \in [0.5000, 1.5000]$ over 60 rad
$\alpha = 2$	$(0.5, -0.9)$, $E^* > 0$	escape	$u = 0$ at $\theta = 0.823$
$\alpha = 4$	$(0.5, 0)$, $E^* = 1/6 < 1/3$	reflected, then escape	$u = 0$ at $\theta = 2.078$
$\alpha = 4$	$(1.2, 0)$, $E^* = 0.288 < 1/3$	collapse	$u \to \infty$ at $\theta = 4.058$
$\alpha = 4$	$(0.5, 0.9)$, $E^* = 0.977 > 1/3$	collapse	$u \to \infty$ at $\theta = 3.604$
$\alpha = 4$	$(0.5, -0.9)$, $E^* = 0.977 > 1/3$	escape	$u = 0$ at $\theta = 0.524$
$\alpha = 4$	$(1, 0)$, $E^* = W_{\max}$	unstable circular orbit	$u = 1$ for 20 rad
$\alpha = 4$	$(1.5, 0)$, $E^* = 0$, $u_0 = u_z$	collapse	$u \to \infty$ at $\theta = 3.139$
$\alpha = 4$	$(0.5, -0.408)$, $E^* = W_{\max}$	escape	$u = 0$ at $\theta = 0.975$
$\alpha = 4$	$(0.5, +0.408)$, $E^* = W_{\max}$	tends to $u_b = 1$ asymptotically	follows $u \to 1$ until round-off, amplified by e^θ, takes it off the separatrix at $\theta \approx 34$

Case	Initial data (u_0, u_0')	Predicted	Integration result
$\alpha = 3$, $\mu/h^2 = 2$	$(1, 0.2)$, $C_1 > 0$	collapse	$u = \cosh\theta + 0.2\sinh\theta$ grows without bound with θ and no zero appears
$\alpha = 3$, $\mu/h^2 = 2$	$(1, -1.5)$, $C_1 < 0$	escape	$u = 0$ at $\theta = 0.805$
$\alpha = 3$, $\mu/h^2 = 1/2$	$(1, 0)$	escape	$u = 0$ at $\theta = 2.221$, equal to $\dfrac{\pi/2}{\sqrt{1/2}}$
$\alpha = 3$, $\mu = h^2$	$(1, -0.1)$	escape at $\theta = 10$	$u = 1 - 0.1\theta$, zero at $\theta = 10$
$\alpha = 1$	$(0.5, 0.3)$	bounded	$u \in [0.471, 1.653]$ over 100 rad

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